

5 (a) (i) Fig. 5.2 B1 [1]

(ii) Fig. 5.3 B1 [1]

(b) kinetic energy increases from zero then decreases to zero B1 [1]

(c) (i) $\Delta E_P = mg\Delta h / mgh$ C1
 $= 94 \times 10^{-3} \times 9.8 \times 2.6 \times 10^{-2}$ using $g = 10$ then -1
 $= 0.024 \text{ J}$ A1 [2]

(ii) either $0.024 = \frac{1}{2} k \times (2.6 \times 10^{-2})^2$ or $\frac{1}{2} kd^2 = \frac{1}{2} k \times (2.6 \times 10^{-2})^2 - \frac{1}{2} kd^2$ C1
 $0.012 = \frac{1}{2} k \times d^2$ $kd^2 = \frac{1}{2} k \times (2.6 \times 10^{-2})^2$ C1
 $d = 0.018 \text{ m}$ $d = 0.018 \text{ m}$
 $= 1.8 \text{ cm}$ $= 1.8 \text{ cm}$ A1 [3]

6 (a) when two (or more) waves meet (at a point) B1